

APX Capabilities



**TRANSACTIONAL
PROCESSING**



**INTEGRATION
WITH SECURITY
SERVICES**



**DATA ACCESS
LAYER**



HOT DEPLOYMENT



**BACKENDS
INTEROPERABILITY**



**BANKING
SERVICES**



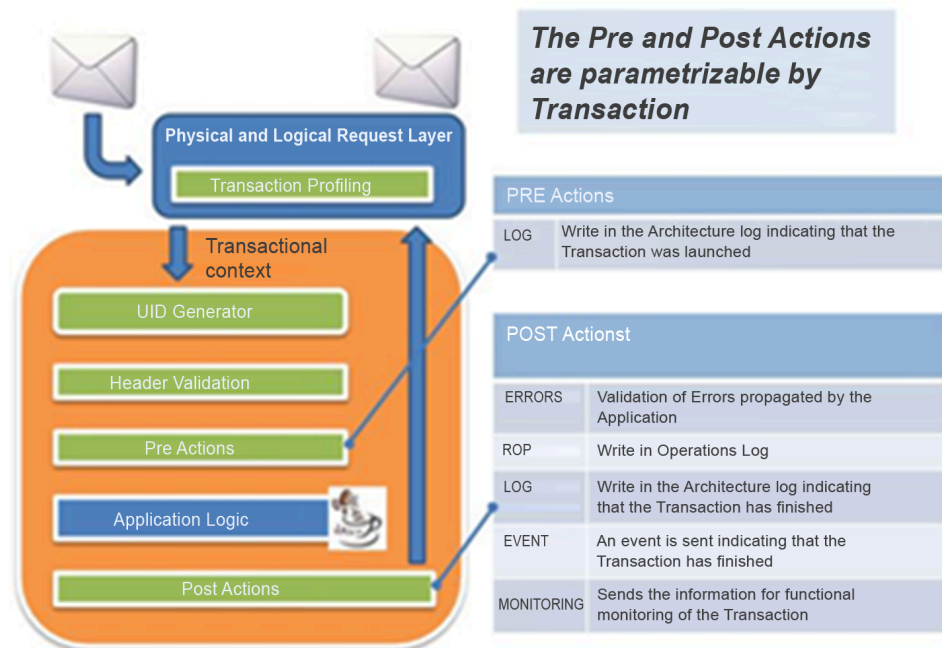
**BATCH
PROCESSING**



MONITORING

Online Architecture execution stack

APX allows creating applications with accessing to relational database and create online transactions processing. The stack of execution in a request to APX is the following:



First of all, APX receives the request in one of the physical protocols that supports (HTTP, JMS, REST,...).

Then create an unique identificator for the transaction and validate if the request headers are well formatted according with the logical protocol defined by Architecture. If everything is ok APX architecture execute the pre-actions, that basically in this point writes the log of execution.

At this point the execution continues in the code that the applications implements for the transaction executing its business logic.

Once application transaction ends the stack executions come back to the control Architecture layer in order to execute the post-actions. These are composed for the following steps:

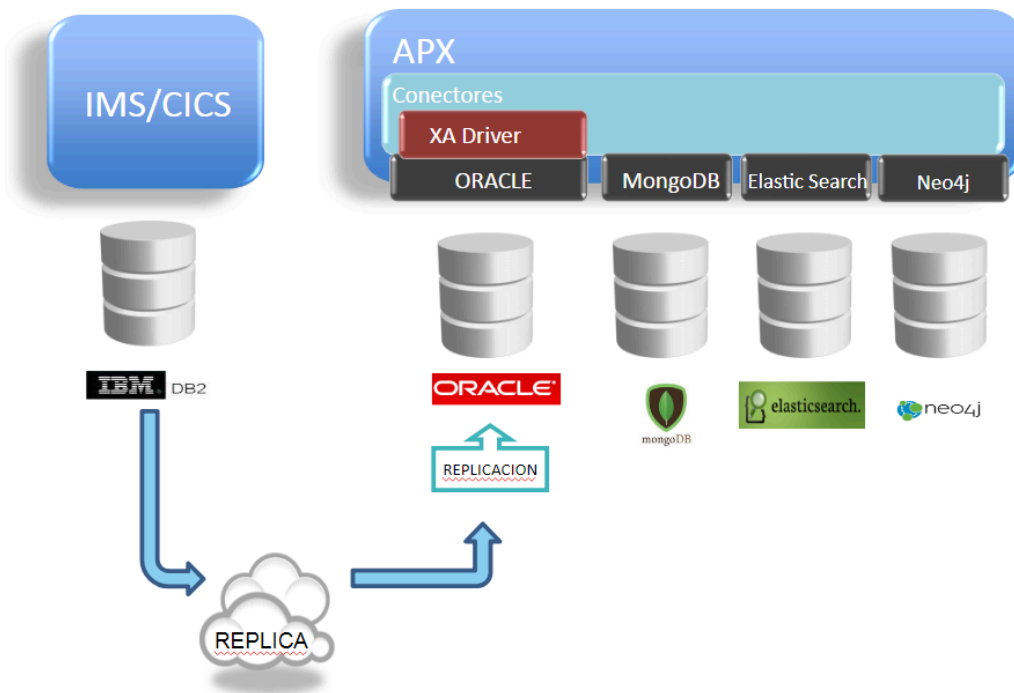
- **Validation of Exit Code:** Architecture retrieves the exit code of the execution and if and error has occurred gets the literal associated.
- **Commit Control:** Architecture controls the commit or Rollback with DB depending of the transaction exit code.
- **ROP Writing.** For each transaction executed, APX registers it in an internal LOG used with Audit, Fraud or Informational purpose indicating that transaction has finished. It is named ROP (Registry Of Operations). This writes is invoked asynchronously.
- **Business Event Notification:** An event with business data is sent when the transaction ends if it is marked for do it. Also this point is executed asynchronously.

- **Monitorization:** During the transaction execution APX is gathered all monitoring information in each steps of the logic implemented by application. In this point the Architecture send it to our Monitoring asynchronously.

Decoupled data access layer

APX Architecture provides to applications the utilities to execute a query through the different kinds of database BBVA supports.

SQL and NOSQL databases are availables to be used by applications in its logical design.

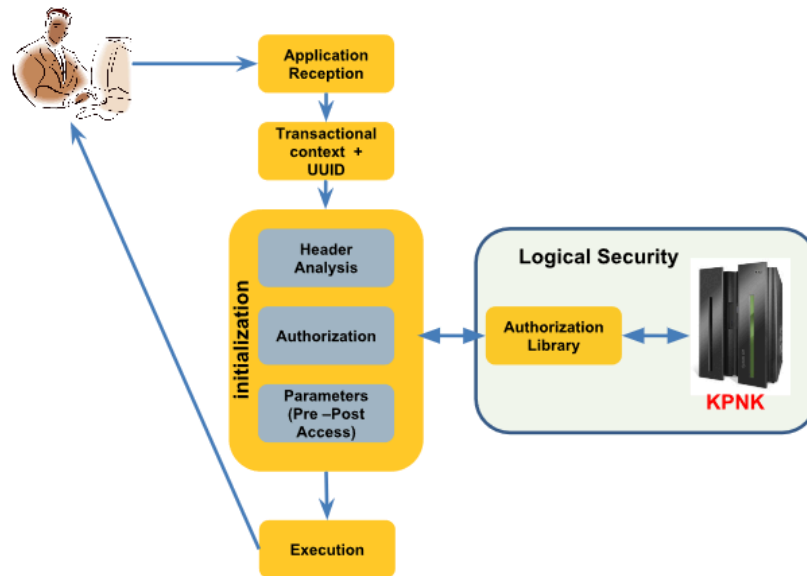


In the case of Oracle, data are accessed through JDBC. The Architecture manages execution of the Commit or Rollback in order to guarantee transactionality.

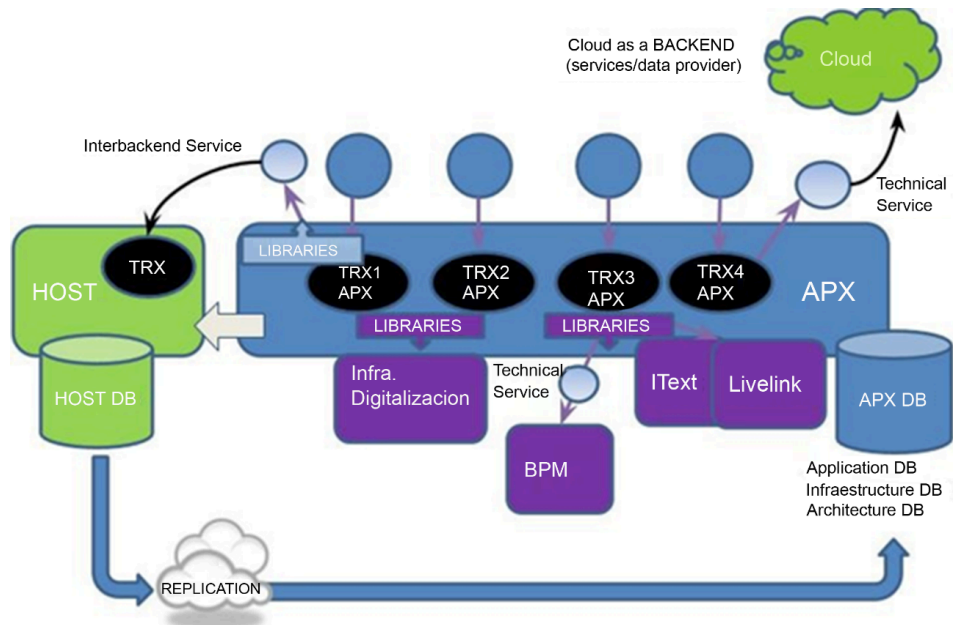
Integration with Corporate Security Services

The request to execute a transaction triggers execution of the authorization process, delegated in Logical Security libraries. The Logical Security components that service the architecture are not coupled to the architecture.

The architecture is independent from implementation of the library.



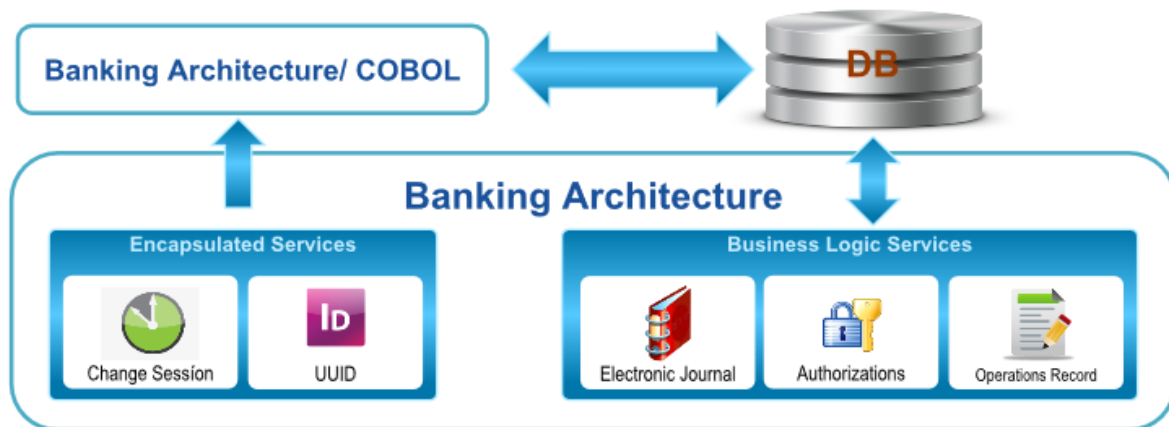
Backend Interoperability



Banking Services adapted to BBVA

APX Arch. provides some of the capabilities of Banking Architecture that PG Architere is providing in Mainframe.

In image below you can see it:



- In the **Session Change** the Architecture detects the change of the Accounting Date from the Host and propagates it.
- **Unique identifier** for transaction/job execution is generated automatically as the first step when the online/batch transactions is invoked. The output of the transaction can be propagated for correlation in the other layers.
- **Electronic Journal** is configured by transaction or UUAA so that the architecture executes a post action after the execution of the transaction. The data model is the same as the Host, allowing to run in standalone or delegate mode (NOT IMPLEMENTED).
- Management of **Authorizations** is delegated in Logical Security modules and Architecture modules (only implemented in Spain for Confidentiality).
- **Operations Registry** is similar to the Audit Trail Host service. The data model is the same as the Host, allowing operation in autonomous or delegated mode.